

No Calm Before the Storm: Typhoons, Shipping, and Trade

Keita Abe^a, Renato Molina^b, Kenta Tanaka^c, Juan Carlos Villaseñor-Derbez^{d,e*}

June 2026

Abstract

Extreme weather is becoming more frequent and severe worldwide, yet its consequences for global trade channels remain an open question. This paper quantifies the cost of extreme weather to maritime trade. Integrating daily typhoon wind-swath data with hundreds of thousands of vessel voyages in Japanese waters, we document that maritime traffic falls by up to roughly 40% on typhoon-exposed open-ocean cells and by about 7% at typhoon-exposed ports. A voyage-level framework identifies \$119 million in direct shipping-industry costs over 2013–2021. This disruption implies a total trade-volume welfare loss of \$117.8 million. Combining the two channels yields a central total welfare estimate of approximately \$237 million. As climate change intensifies, the invisible costs associated with increased extreme weather will scale accordingly, highlighting a vital dimension of climate risk that remains overlooked by adaptation frameworks focused solely on catastrophic land-based destruction.

JEL: F14, F18, Q54, R41

Keywords: typhoons, maritime shipping, trade costs, natural disasters, Japan, AIS data

^{*a}Keio University, Minato City, Tokyo, Japan. ^bDepartment of Economics, Appalachian State University, Boone, North Carolina, USA. ^cMusashi University, Nerima City, Tokyo, Japan. ^dDepartment of Environmental Science and Policy, Rosenstiel School of Marine, Atmospheric & Earth Science, University of Miami, Miami, Florida, USA. ^eFrost Institute for Data Science & Computing, University of Miami, Miami, Florida, USA. Corresponding author: molinar@appstate.edu. We thank seminar and conference participants at J-TREE for helpful comments.

1 Introduction

Over 80% of global trade by volume moves by sea (UNCTAD, 2021), and the Western Pacific, the world’s most active tropical cyclone basin, sits astride the shipping lanes that connect Asia’s largest economies. These two facts place an enormous and growing share of world commerce at the mercy of tropical cyclones in the Pacific, also known as typhoons. The problem is worsening. Typhoon intensity has nearly doubled since the 1970s (Emanuel, 2005), and climate projections indicate further increases in the proportion of storms reaching the most destructive categories (Knutson et al., 2020). Yet despite this trajectory, evidence on how much tropical cyclones cost the shipping industry, or how those costs propagate to trade, remains limited.

Here, we tackle this problem focusing on Japan, the economy most exposed to this risk. Japan is the world’s fourth-largest trading nation, with merchandise trade exceeding \$1.4 trillion in 2021, and over 99% of its volume moves on ships. Unlike continental economies with land-border alternatives, Japan has no fallback: it imports virtually all of its crude oil and liquefied natural gas (IEA, 2021), over 60% of its food calories (MAFF, 2022), and the bulk of its manufacturing inputs entirely by sea. Beyond its links to North and South America, its ports are connected to trading partners in China, Southeast Asia, Australia, and the Middle East by routes that thread directly through the Western North Pacific, the world’s most active tropical cyclone basin.

Using Automatic Identification System (AIS) signals to reconstruct vessel trajectories globally, we estimate that traffic falls by up to roughly 40% in typhoon-exposed ocean areas and by about 7% at affected ports. The open-ocean response is sharp because underway vessels can still reroute or delay; the port-level response is muted because vessels arriving at port are largely committed and most ride out the storm in place. The effects are concentrated in the July–October typhoon season, recover within three to five days, and strengthen with storm intensity, which lengthens the exposure window through which costs accrue.

With this evidence of avoidance by vessels, we then quantify the cost through two chan-

nels. The first is the direct shipping-industry cost, which accrues via adjustments in fuel and labor costs associated with rerouting. Using a voyage-level panel of over 700 thousand international trips relevant to Japan, we compare each typhoon-exposed voyage to matched untreated voyages on the same route, vessel type, and year. Our estimates indicate that the direct cost is \$119 million over 2013–2021, a 1.2% penalty on the 59,219 treated voyages in our sample.

The second channel is a trade-volume welfare loss. We embed the voyage-level transport-cost shock in a bilateral gravity model (Anderson and van Wincoop, 2003), instrumenting bilateral shipping cost with the share of voyages exposed to active typhoon wind swaths in the corresponding month. The estimated short-run trade-cost elasticity is $\widehat{\sigma - 1} \approx 1.29$, implying an additional \$117.8 million in trade-volume welfare loss at the point estimate. Together, the two channels yield a central total of roughly \$237 million over 2013–2021 (about \$26 million per year).

This paper makes three primary contributions to the literature. First, we advance the literature on the macroeconomic and trade impacts of natural disasters. Previous work has largely evaluated disasters through the lens of post-landfall capital destruction, aggregate output shocks, and fiscal transfers (Hsiang and Jina, 2014; Strobl, 2011; Felbermayr and Gröschl, 2014; Deryugina, 2017; Pelli and Tschopp, 2017; Dell et al., 2014; Krichene et al., 2023), or traced how these terrestrial damages propagate through firm-level supply chains (Boehm et al., 2019; Carvalho et al., 2021; Barrot and Sauvagnat, 2016). This body of work identifies disaster costs almost entirely from spatially aggregated outcomes such as national or regional output and ex-post damage tallies, an approach whose ability to recover the specific channels through which damage operates has recently been questioned (Schötz, 2025). In this context, we isolate an entirely distinct, pre-landfall mechanism: the disruption of the maritime transport link itself. By measuring the avoidance behavior of vessels while storms are still at sea, we capture economic losses that occur even when coastal infrastructure and downstream production facilities remain completely intact, and that are invisible to the

aggregate accounts on which the macro literature relies.

Second, we contribute to the empirical literature on the structural costs of shipping disruptions. While recent studies have used observed vessel detours and reroutings to quantify the welfare burden of various maritime hazards such as piracy in major shipping lanes (Bensassi and Martínez-Zarzoso, 2013; Vespe et al., 2015; Besley et al., 2015; Molina et al., 2024), marine-resource enforcement at no-take boundaries (Englander, 2019), and historic choke-point closures (Feyrer, 2021), we extend this framework to transient, high-frequency weather shocks. A nascent transport-science literature has begun to use AIS data to document how individual tropical cyclones disrupt vessel movements and shipping networks (Verschuur et al., 2020; Zhang et al., 2024; Huang et al., 2024), but these are descriptive single-storm or single-port case studies. Here, we assemble a multi-year, fleet-wide voyage panel and translate the disruption into monetized industry costs and a structural welfare loss. This approach allows us to build a consistent micro-to-macro accounting of weather-driven trade disruption that captures the complex margins of speed, timing, and routing adjustments.

Finally, we contribute to the literature on maritime transport costs and structural gravity by estimating the short-run trade-cost elasticity using a multi-segment Asia-Pacific fleet under weather variation. The standard gravity framework relies heavily on cross-sectional distance to identify the trade elasticity, yielding a long-run equilibrium benchmark that reflects decades of adjustment to permanent cost differentials (Anderson and van Wincoop, 2004; Limão and Venables, 2001; Head and Mayer, 2014). The recent IV-based shipping-cost literature (Brancaccio et al., 2020; Asturias, 2020; Wong, 2022) identifies the same elasticity from supply-side instruments that shift transport prices but not bilateral trade demand directly, and consistently finds short-run elasticities far below the cross-sectional benchmark. For example, Brancaccio et al. (2020) report -1.03 for dry bulk using tariff instruments; Wong (2022) and Asturias (2020) report -3 to -5 for container shipping. We extend this literature by using transient typhoon exposure as a novel supply-side instrument for bilateral shipping costs. Typhoon timing is plausibly exogenous to global monthly bilateral

trade demand, which we believe provides a cleaner shifter than the tariff and population instruments used in prior work. The recovered elasticity here can thus be interpreted as the short-run analog of the long-run benchmark, which we believe is also the appropriate object for evaluating the welfare costs of transient climate-induced shocks.

The paper proceeds as follows. Section 2 provides additional background on Japan’s maritime trade, the port system, and typhoon climatology. Section 3 describes the data sources. Section 4 presents the empirical strategy: the exposure measure, the event-study design, the voyage-level matched-control framework, and the structural gravity model. Section 5 reports results, and Section 6 discusses scope, interpretation, and policy implications. Section 7 concludes.

2 Background

Japan is an island economy whose prosperity depends on the sea. Total merchandise trade exceeded \$1.4 trillion in 2021, and over 99% of it by volume moved on ships (MLIT, 2022).¹ The dependence is especially acute for energy and food. Japan imports virtually all of its crude oil (mostly from the Middle East via tanker through the Strait of Malacca), all of its liquefied natural gas (from Australia, Qatar, and Southeast Asia), and over 60% of its food calories. Its top trading partners (i.e., China, the United States, Australia, South Korea, and the ASEAN economies) are all connected by ocean routes that traverse the Western Pacific. Any disruption to maritime transport thus has immediate consequences for Japan’s energy supply, food security, and manufacturing inputs.

Relatedly, the Western Pacific is the most active tropical cyclone basin in the world, generating an average of 26 named storms per year, which is roughly twice the Atlantic average (Schreck III et al., 2014). Of these, 10 to 12 approach the Japanese archipelago and three to five make landfall, typically between July and October. Storms that form in the

¹By value, the sea share is lower because multiple high-value electronics and pharmaceuticals travel by air, but even by value maritime transport accounts for roughly 70% of Japan’s total trade.

tropical Pacific near the Mariana Islands tend to recurve northeastward as they encounter the mid-latitude westerlies, placing Japan’s southern and central coasts, and the shipping lanes that connect them to East and Southeast Asia, squarely in the path of recurring typhoons.

The 2018 and 2019 seasons were particularly destructive. Typhoon Jebi (September 2018) was the strongest typhoon to make landfall in Japan in 25 years (Mori et al., 2019), closing Kansai International Airport and the Port of Kobe for days. Typhoon Faxai (September 2019) struck the Tokyo Bay area with sustained winds exceeding 150 km/h, disrupting port operations across the Keihin complex. Typhoon Hagibis (October 2019), one of the largest storms on record in the basin, caused widespread flooding across eastern Japan. Importantly, we note that intensity in the basin has been rising over recent decades (Emanuel, 2005; Knutson et al., 2020), a trend that climate projections suggest will continue.

For the shipping industry, the relevant footprint of a typhoon is not necessarily landfall damage but the wind swath over open water. A typhoon’s 30-knot (55.6 km/h) and 50-knot (92.6 km/h) wind radii can extend 300 to 500 kilometers from the storm center, covering vast areas of ocean for two to four days as the storm transits (Sampson et al., 2017).² As a response, captains may slow down to let the storm pass ahead, adding hours or days to the voyage and increasing labor costs. They may also reroute to avoid the swath entirely, adding distance and fuel cost. They may delay departure from port, waiting for the storm to clear. Port authorities, for their part, may restrict vessel entry, close berths to prevent damage from surge and high winds, or suspend cargo operations.

These responses differ from landfall damage in an important way: they are largely preventive. The ship is not destroyed and the port infrastructure is not wrecked (absent a massive catastrophe). But the disruption to schedules, fuel budgets, and cargo delivery is real and costly, and it likely accumulates over a typhoon season with 10 to 15 storms.

Against this background, the fleet serving Japanese international trade is also heteroge-

²A wind radius is the maximum distance from the storm center at which a given sustained-wind threshold is observed. The wind field is typically asymmetric, so radii are reported separately for the four quadrants of the storm; we use the long and short sides directly to construct the four daily wind-swath polygons described in Section 4.1.

neous, and the cost of typhoon disruption is likely to vary across vessel types. Bulk carriers carrying iron ore, coal, and grain operate on “tramp” schedules, sailing when a full cargo is available (Brancaccio et al., 2020). Their schedules are flexible, and a one-day delay is relatively inexpensive. Tankers carry crude oil and petroleum products on both spot and time-charter arrangements (Stopford, 2009); their fuel consumption is high (a large crude carrier burns 60–80 metric tons of heavy fuel oil per day at sea), so rerouting is costly even if schedule flexibility is moderate. Container ships operate on fixed liner schedules with published port calls and tight connections to inland logistics chains. A delay of even a few hours can cascade through the supply chain, making container traffic the most schedule-sensitive segment (Hummels, 2007; Hummels and Schaur, 2013; Bernhofen et al., 2016).³ Our analysis explores the burden of extreme weather across these different vessels explicitly.

3 Data

Our analysis focuses on traffic across space, vessel voyages, and the bilateral port-month trade. To quantify the economic cost of maritime weather disruptions then, we will use data on typhoon hazard exposure, vessel movements, voyage-level costs, and bilateral trade flows.

3.1 Typhoons and Shipping Routes

We reconstruct the spatial footprint of typhoons using hourly track data from the National Institute of Informatics’ Digital Typhoon database (Kitamoto et al., 2023). This dataset provides center coordinates, central pressure, maximum sustained winds, and wind radii for both 30-knot and 50-knot thresholds. Because storm wind fields are typically asymmetrical, these radii are reported separately for the long and short sides of the storm. We buffer the center positions by these radii to construct four distinct daily wind swath polygons per

³Car carriers and refrigerated vessels (reefers) are also present in the fleet but account for a smaller share of total voyages. Car carriers serving Japanese auto exports to North America and Europe operate on semi-regular schedules.

typhoon (see Section 4.1). Figure 1 illustrates this construction for Typhoon Jebi (2018) on September 4, as the storm made landfall over Japan.

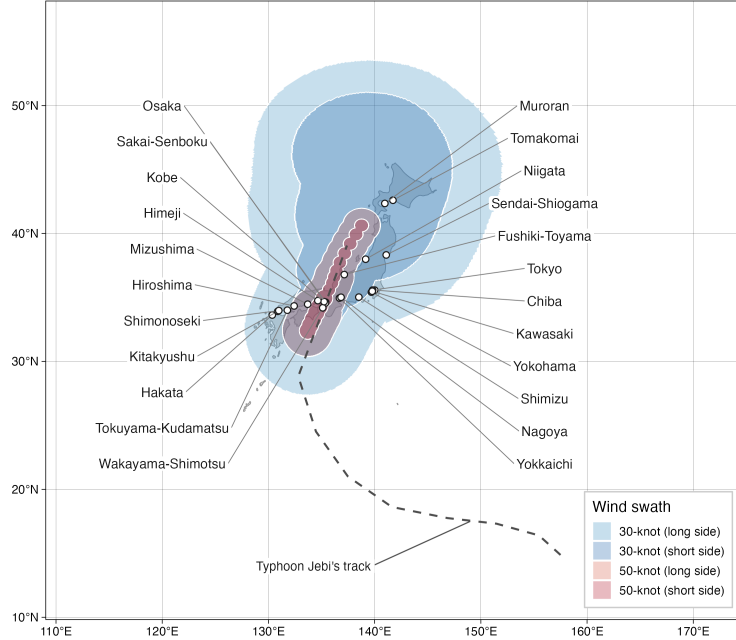


Figure 1. Wind-swath construction for Typhoon Jebi (2018) on September 4, 2018. Each polygon is the union of circular buffers placed at the storm’s 6-hourly center positions, with radius equal to the reported wind radius at the given threshold and side of the storm: light blue = 30-knot long side; dark blue = 30-knot short side; light red = 50-knot long side; dark red = 50-knot short side. The dashed line shows the storm track leading up to September 4. Open circles mark the internationally important Japanese ports (MLIT classes 11–12) used in the direct and indirect port-level analysis.

To link these typhoons to maritime exposure, we construct least-cost maritime paths between observed origin-destination pairs on a 1° conductance raster cropped to 60°S – 70°N latitude to avoid routing through the Arctic. The algorithm connects centroids of all 1° cells that make up the shortest path along the conductance raster. We restrict this analysis to routes for Japanese ports of class-11/12/13 that connect with foreign ports and that had at least 10 recorded trips. Foreign ports outside the 60°S – 70°N are also removed to prevent unrealistic Arctic routing.

Coordinates for Japanese ports come from the National Land Numerical Information

Port and Harbour dataset (Ministry of Land, Infrastructure, Transport and Tourism, 2014). For foreign ports, we use the centroid of all anchorages associated with each port based on data provided by Global Fishing Watch (2026). The resulting shipping network connects these Japanese ports (with Tokyo Bay ports sharing Yokohama’s route geometries) to 540 foreign ports across 54 countries, yielding 4,360 bilateral routes.

Daily swath exposure per route is then pre-computed by geometric intersection, giving us a balanced route-day panel with binary indicators. Figure 2 shows the resulting network. Appendix A.6 maps the spatial distribution of typhoon exposure across the network and the bilateral cells that contribute most of the identifying variation.

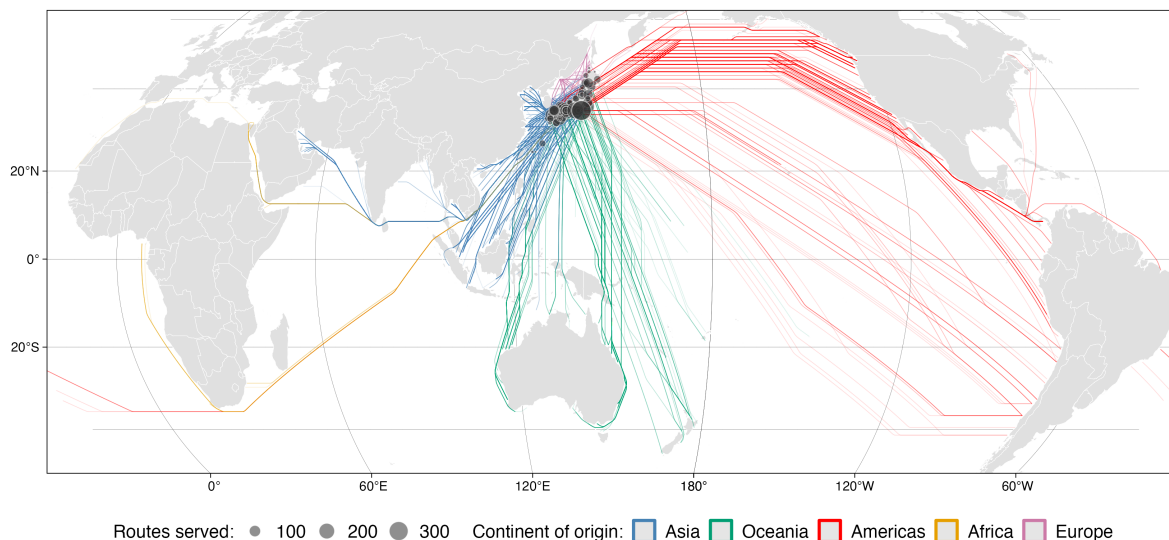


Figure 2. International shipping routes connecting Japanese ports to foreign ports in 47 countries. Each line is a least-cost ocean path computed on a 1° conductance raster with a 16-direction transition matrix, colored by the continent of the foreign origin: Asia, Oceania, Americas, Africa, or Europe. Filled circles mark the Japanese ports, sized by the number of routes each serves.

3.2 Vessel Movements and Voyage Costs

We obtain data on vessel tracks and maritime activity from Global Fishing Watch (GFW), which processes high-resolution timestamped latitude–longitude messages from the Auto-

matic Identification System (AIS) received via satellite, terrestrial, and dynamic receivers (Kroodsma et al., 2018). AIS transponders are required on vessels above 300 gross registered tons operating internationally and by many countries within exclusive economic zones (International Maritime Organization, 2000). We restrict the panel to vessels classified as cargo, tanker, reefer, bunker, or one of their hybrid categories (cargo or tanker, bunker or tanker, cargo or reefer, specialized reefer, container reefer).

We utilize the AIS data in two forms: (i) a daily 0.5° gridded transit panel that records vessel hours, distance, and trip counts per cell-day for all international trips originating or ending in Japan; and (ii) a voyage-level dataset constructed by aggregating raw AIS pings into port-to-port trips using GFW’s port-visit algorithm (Global Fishing Watch, 2021). A voyage is defined as a single trip between two consecutive port calls. Multi-segment journeys are treated as a series of individual voyages.

We filter these data to include only voyages with reliable profiles. That is, total distance and duration each positive, total duration at most 60 days, total distance at most the earth’s circumference (40,075 km) and at most four times the average observed distance for the port-to-port combination, known origin and destination countries, observed distance at least as large as the great-circle distance between the start and end ports, and an estimated design speed above 10 knots. Restricting to international voyages departing from or arriving at Japanese ports between 2013 and 2021 yields a panel of 536,380 valid voyages across 19,598 unique vessels.

For each voyage, we then calculate operational cost as the sum of fuel and labor cost. Fuel consumption is computed from vessel main-engine power, gross tonnage, auxiliary-engine power, and design speed. Main-engine power and gross tonnage come from the GFW vessel-characteristics database (Kroodsma et al., 2018), populated first from registered information and then, where missing, from the GFW vessel-characteristic neural network. Auxiliary power is derived from main-engine power using the empirical relationships of Betz (2011), which link main propulsion requirements to auxiliary needs. Design speed is computed as a

function of main-engine power and gross tonnage following Betz (2011). However, because that regression occasionally yields implausibly low values, we restrict the voyage panel to vessels with estimated design speed above 10 knots. We then calculate fuel consumption following a standard engineering approach (Corbett et al., 2009) that combines main-engine and auxiliary-engine consumption. Main-engine fuel consumption is the product of operating hours, main-engine power, the main-engine specific fuel-consumption rate (Wang et al., 2007), and a cubic law of operating speed relative to design speed. Auxiliary consumption is the product of operating hours, auxiliary-engine power, and the auxiliary-engine specific fuel-consumption rate (Wang et al., 2007). Fuel consumption is finally calculated at the AIS-ping level and summed across pings within a voyage.

Daily fuel-price data come from the Bunker Index 380 CST series (Bunker Index, 2026). This is the global average price across all ports selling 380 centistoke fuel, the most commonly used grade in maritime transport. Bunker prices are reported on business days. We forward-fill missing daily values from the most recent reported price (most gaps span no more than two days). Total fuel cost for each voyage is the product of total fuel consumption and the bunker price on the voyage’s departure date. Labor cost is computed by combining a vessel-size- and type-specific crew estimate from Betz (2011) with the 2018 International Transport Workers’ Federation wage scale for the average non-officer seafarer (International Transport Workers’ Federation, 2025). Appendix A.1 reports the pooled unit-cost summaries from our voyage panel (\$38.71/nm fuel, \$3,354/day labor) and the per-vessel-type and per-tonnage-decile breakdowns.

3.3 Customs Trade and Bunker Prices

Data on bilateral trade flows come from the Trade Statistics of Japan. We collect monthly export and import values by port for the primary class-11/12 ports (International Strategic and Hub Ports) between 2013 and 2021; the class-13 ports in our shipping network appear in vessel movements but not in port-level customs records. We convert them to US dollars using

the monthly average JPY/USD spot rate from FRED (series EXJPUS) (Board of Governors of the Federal Reserve System , US).

The bilateral panel used in the gravity model in Section 4.4 is restricted to port-country-month cells with realized customs imports above zero. This yields 56,091 observations covering 39 origin countries on 698 active port-country pairs.

4 Empirical Strategy

4.1 Measuring Typhoon Exposure

To quantify the spatial and temporal footprint of each typhoon, we rely on the reported wind radii at both 30- and 50-knot thresholds. We construct three specific measures of maritime exposure. First, at the grid-cell level, we rasterize the swaths over a 0.5° grid covering Japan’s Exclusive Economic Zone (EEZ), creating a binary exposure panel that spans approximately 1,357 active cells over 3,288 days to match our transit data.

Second, at the direct port level, we define a port as exposed on a given day if its coordinates fall within any active swath polygon, allowing us to track daily vessel departures and arrivals for the primary class-11/12 ports. Third and finally, we construct an indirect-port exposure panel that defines a shipping route as exposed if its geometry intersects an active swath, aggregating route-level exposure to the port-day level using both unweighted and trip-volume-weighted route shares.

4.2 Event-Study Design

To capture the dynamic traffic responses to these exposures, we implement an event-study design. A unit (cell or port) is considered treated on a given day if it experiences at least 30-knot wind exposure. For each exposure panel, we group exposure days into treatment episodes within a 14-day window and assign relative-time indicators. We estimate the fol-

lowing specification:

$$\text{asinh}(Y_{it}) = \sum_{k=-K}^K \beta_k \cdot \mathbf{1}[\text{relative time}_{it} = k] + \alpha_i + \gamma_t + \varepsilon_{it} \quad (1)$$

where Y_{it} represents the maritime traffic outcome (e.g., vessel hours, port departures) transformed via the inverse hyperbolic sine, α_i and γ_t are unit and time fixed effects, respectively, and $k = -7$ serves as the omitted reference period. The identification assumption is that the timing of a typhoon is exogenous to contemporaneous shocks to maritime trade, after controlling for the set of unit and time fixed effects. We calculate standard errors following Conley (1999), applying a 100-km spatial distance cutoff to account for spatial correlation.⁴

4.3 Structural Cost Quantification

Next, we estimate the direct economic costs incurred by vessels navigating these weather shocks. We define a voyage as treated if its least-cost route intersects any active wind swath polygon during its specific sailing window. To isolate the excess cost of this exposure, we compare each treated voyage to a set of matched, untreated control voyages. Specifically, we calculate the median cost per optimal-route nautical mile among untreated voyages matched on the same origin-destination route, vessel type, and year, where “optimal-route distance” is the length of the least-cost maritime path between the two ports (Section 3). If the tightest matching stratum lacks at least five control voyages, we progressively coarsen the match to route-by-vessel-type, and finally to route-only.⁵ The expected cost for a treated voyage is then defined as this matched cell median multiplied by the voyage’s optimal-route distance, with the excess cost being the simple difference between realized and expected costs.

⁴The recent literature on staggered event-study estimators (Sun and Abraham, 2021) highlights bias from heterogeneous treatment effects when treatment timing varies across units and the absorbing-state assumption fails. Those concerns do not bind here: typhoon exposure is binary and transient, the same unit is repeatedly exposed across the panel, and there is no absorbing treatment state to weight inconsistently.

⁵Vessel age is not directly observable in the GFW AIS panel; we use tonnage decile (entered as a fixed effect in the parametric corroboration in Table 1) as a proxy for vessel size and engineering vintage. To the extent that vessel age varies independently of route, vessel type, and year within a stratum, its omission attenuates rather than biases the matched-control comparison.

A key empirical challenge in this matching process is the seasonal clustering of typhoons. Because treated voyages are concentrated between July and October while control voyages span the entire year, intra-year volatility in global bunker fuel prices (e.g., a 42% drop between May and December in 2015) can confound the fuel cost comparison. To address this, we match voyages based on fuel intensity rather than raw fuel cost:

$$k_i = \frac{\text{fuel cost}_i}{\ell_i \times P_i^{\text{bunker}}}, \quad (2)$$

where ℓ_i is the optimal-route distance for voyage i 's origin-destination pair and k_i captures the fuel consumed per nautical mile per dollar of bunker fuel, which is vessel-specific and invariant to the bunker price cycle. The expected fuel cost for a treated voyage is then calculated as the median fuel intensity of the control group multiplied by the treated voyage's optimal-route distance and its prevailing bunker price. For labor costs, we maintain the raw cost-per-route-nm metric, as crew wages do not scale with global bunker price fluctuations.

4.4 Bilateral Trade-Cost Elasticity

To translate the voyage-level transport-cost shock into a structural trade response, we embed it within an Anderson–van Wincoop (Anderson and van Wincoop, 2003) CES gravity framework. Bilateral imports from country c to port j in month t are given by:

$$X_{jct} = \frac{Y_{ct} E_{jt}}{Y_t} \left(\frac{\tau_{jct}}{\Pi_{ct} P_{jt}} \right)^{1-\sigma}. \quad (3)$$

X_{jct} is the value of imports from country c entering through port j in month t , the dependent variable we observe in the customs panel. Y_{ct} is country c 's total exports in month t (origin output), E_{jt} is total expenditure on imports at port j in month t , and Y_t is aggregate world output in month t , which normalizes the share of country- c supply that reaches a representative destination. τ_{jct} is the bilateral iceberg trade cost on the j – c route in month t , i.e., the resource cost of moving a unit of goods from c to j , of which our

voyage-level shipping cost is the typhoon-responsive component.

Π_{ct} and P_{jt} are the outward and inward multilateral resistance terms: Π_{ct} is a price-index-weighted average of country c 's trade costs to all destinations (capturing competition for c 's supply), and P_{jt} is a price-index-weighted average of port j 's trade costs from all origins (capturing competition for j 's expenditure). Finally, $\sigma > 1$ is the elasticity of substitution across origin countries in the CES demand system, so $\sigma - 1$ is the trade elasticity (Head and Mayer, 2014). Taking logs and writing the bilateral transport cost as a port-country mean cost per voyage, $\bar{\tau}_{jct}$, yields our estimating equation:

$$\log X_{jct} = \alpha_{jc} + \phi_{jt} + \psi_{ct} - (\sigma - 1) \log \bar{\tau}_{jct} + u_{jct}, \quad (4)$$

where α_{jc} is a port-country pair fixed effect that absorbs time-invariant bilateral fundamentals (distance, language, tariff schedules, long-run trade relationships); ϕ_{jt} is a port-month fixed effect that absorbs the inward multilateral resistance P_{jt} together with any other port-by-month variation in absorption. ψ_{ct} is a country-month fixed effect that absorbs the outward multilateral resistance Π_{ct} together with origin-month supply shocks; and u_{jct} is an idiosyncratic disturbance. The coefficient on $\log \bar{\tau}_{jct}$ identifies $-(\sigma - 1)$, the trade elasticity expressed as a negative quantity because higher transport cost reduces bilateral imports.

It follows that simple OLS estimation of equation (4) would yield biased estimates of $\sigma - 1$ because the bilateral shipping cost is determined jointly with trade flows in equilibrium. In other words, ships' bargaining position depends on the destination's loading prospects, which themselves depend on trade flows everywhere. We address this endogeneity by instrumenting $\log \bar{\tau}_{jct}$ with typhoon-exposure measures on the bilateral cell. Our preferred instruments are two shares of voyages on route jc in month t that overlap an active wind swath, decomposed by storm intensity: the 50-knot share Z_{jct}^{50} and the 30-knot-only share Z_{jct}^{30} (i.e., voyages exposed to a 30-knot perimeter that does not also have a 50-knot core). Each enters the first-stage regression:

$$\log \bar{\tau}_{jct} = \pi_{jc} + \chi_t + \gamma_{50} Z_{jct}^{50} + \gamma_{30} Z_{jct}^{30} + v_{jct}, \quad (5)$$

where π_{jc} is a port-country pair fixed effect and χ_t is a year fixed effect. The exclusion restriction is that monthly typhoon exposure on a jc route shifts the cost of operating that route but is otherwise exogenous to bilateral trade demand at the port-country-month level, conditional on these fixed effects.⁶ Typhoon timing follows meteorological and climatic patterns, and varies idiosyncratically across years and routes within a year. This provides variation that is plausibly exogenous to the bilateral fundamentals absorbed by π_{jc} and the aggregate annual shocks absorbed by χ_t . We cluster standard errors at the pair $\times$ year level, the granularity at which typhoon exposure is assigned to a given route over the panel. Appendix A.4 reports robustness across alternative specifications.

5 Results

5.1 Reduced-form evidence of the mechanism: traffic disruption

We begin with reduced-form evidence that typhoon exposure causes large, sharp, and transient disruptions to maritime traffic. Figure 3 presents the grid-cell event study. Each panel plots the coefficient $\hat{\beta}_k$ from equation (1) for one traffic outcome (transits, time in cell, distance traveled).

The results are coordinated across the three cell-day aggregates. On the day a typhoon’s wind swath covers an ocean grid cell, the count of vessel transits through the cell falls about 23%, total vessel-hours about 28%, and total vessel-distance about 40% relative to the pre-event baseline seven days before exposure.⁷ A similar response occurs one day earlier, which

⁶A standard concern with weather-based instruments is that extreme events might violate this restriction by physically destroying destination port infrastructure, thereby directly halting trade. Our results below show that this is not a concern, as port disruptions in our sample are transient operational delays rather than structural capital destruction.

⁷Because the traffic outcomes are measured in inverse hyperbolic sine, we convert each event-study coefficient β into a proportional change as $100(e^\beta - 1)\%$, the semi-elasticity for asinh-transformed outcomes

is consistent with captains acting on the typhoon forecast to reroute around the affected area or delay departure.

The ordering of the three aggregate magnitudes (distance > time > transits) is consistent with avoidance behavior operating on both the extensive margin (fewer vessels enter the cell) and the intensive margin (vessels that do enter cover less ground inside it). The cell-level data cannot separately identify these two channels, but we tackle this issue directly in Section 4.3 below. After exposure clears, traffic rebounds, exceeds the baseline for 3 days, and returns to baseline in about five days. The additional traffic observed in the days after exposure is consistent with behavior of vessels that delayed their arrival/departure while waiting for the storm to pass. Appendix A.7 reports the full pipeline replicated with treatment restricted to the inner 50-knot wind core.

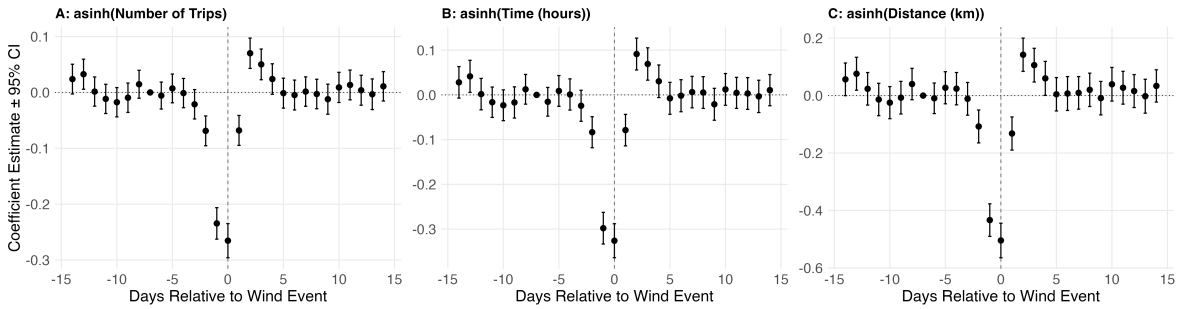


Figure 3. Grid-cell event study: effect of typhoon exposure on vessel transits, time in cell, and distance traveled. Points are coefficient estimates $\hat{\beta}_k$ from equation (1). Outcomes are in inverse hyperbolic sine. Treatment is exposure to at least 30-knot winds. Reference period is $k = -7$. Whiskers are 95% confidence intervals based on Conley spatial HAC standard errors (100 km distance cutoff).

Figure 4 presents the parallel analysis at the port level. The decline is much more muted than at the cell level. Vessel departures fall by about 4% (though not significant), arrivals by about 5%, and total trips by about 7% during the day of exposure, with an anticipatory drop of roughly the same magnitude the day before. The smaller magnitude is expected as ports are fixed destinations, so vessels already docked or approaching simply pause operations or wait at anchorage rather than rerouting. Consequently, these brief operational delays

away from zero.

are smoothed across the daily aggregate, with the bulk of the financial burden accruing during the open-ocean avoidance phase. Recovery is similarly rapid, with traffic returning to baseline within three to five days, and the pattern is consistent across departures, arrivals, and total trips.

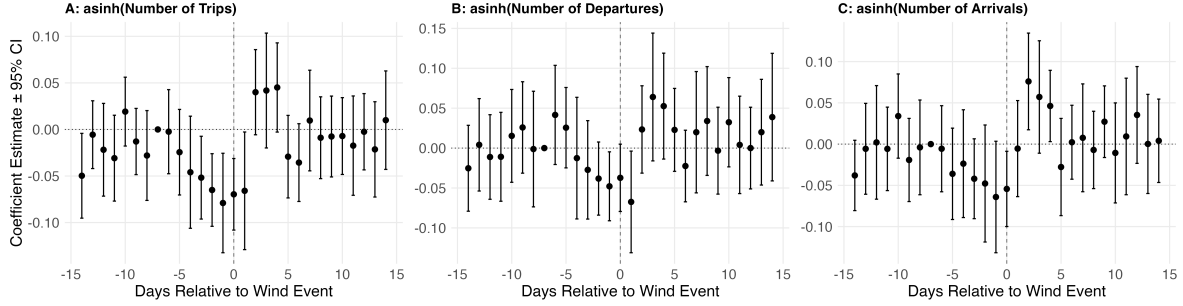


Figure 4. Port-level event study: effect of typhoon exposure on vessel trips, departures, and arrivals. Points are coefficient estimates $\hat{\beta}_k$ from equation (1). Outcomes are in inverse hyperbolic sine. Treatment is exposure to at least 30-knot winds. Reference period is $k = -7$. Whiskers are 95% confidence intervals based on Conley spatial HAC standard errors (100 km distance cutoff).

5.2 Direct Shipping Cost Penalty

We next translate the reduced-form traffic effects into the dollar cost of typhoon exposure at the voyage level. The matched-control estimation described in Section 4.3 covers 59,219 treated voyages. These are international trips whose least-cost route intersected an active wind swath during the sailing window. Of these, 75% match at the tightest stratum (route $\times$ vessel type $\times$ year), ensuring that the counterfactual cost comes from voyages on the same route, in the same type of vessel, facing the same annual bunker and labor market conditions. A further 21% match at the intermediate stratum (route $\times$ vessel type), 2% match at the route-only stratum, and the remaining 2% receive a thin-cell route-mean fallback. Appendix A.2 reports balance diagnostics across treated and matched-control voyages on observable pre-treatment characteristics (route length, vessel tonnage, design speed, departure year).

The total excess cost is \$119 million over 2013–2021, representing a 1.2% cost penalty on treated voyages, or roughly \$13 million per year. To corroborate this matched-control

accounting, Table 1 reports voyage-level OLS regressions of fuel, labor, and total cost on a binary typhoon-exposure indicator. Standard errors are clustered at the year-month level to account for the common-shock nature of typhoons.

The operating-cost penalty is sharply identified in labor where treated voyages incur \$2,000 more in labor cost per voyage, which is consistent with extended time at sea. Fuel cost, by contrast, is highly variable across storms and its coefficient is statistically indistinguishable from zero. The total-cost coefficient, \$2,400 per voyage, is correspondingly positive but inherits the fuel-side volatility. The regression thus corroborates the matched-control mechanism: the cost of typhoon avoidance accrues through additional time at sea rather than additional fuel burned.

Table 1. Voyage-level treatment effects on operating cost.

	Fuel cost (USD) Fuel cost (1)	Labor cost (USD) Labor cost (2)	Total cost (USD) Total cost (3)
Typhoon-exposed	442.4 (2,996.4)	1,989.0*** (127.5)	2,431.4 (2,985.5)
Observations	441,932	441,932	441,932
R ²	0.81603	0.88870	0.85510
Route fixed effects	✓	✓	✓
Vessel type fixed effects	✓	✓	✓
Tonnage decile fixed effects	✓	✓	✓
Year-month fixed effects	✓	✓	✓

Notes: Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. OLS regressions of voyage-level fuel cost (column 1), labor cost (column 2), and total operating cost (column 3) on a binary typhoon-exposure indicator. Sample is the matched-voyage panel from Section 4.3. Fixed effects: bilateral route, vessel type, tonnage decile, and year-month. Standard errors clustered at the year-month level.

Table 2 breaks the cost down by year. The annual pattern closely tracks the severity of the typhoon season. The costliest year in our sample is 2018, at \$29 million, dominated by Typhoon Jebi, which made landfall near Kobe in September as the strongest typhoon to hit Japan in 25 years. The second-costliest year is 2019, at \$28 million, driven by the two exceptionally powerful storms Faxai and Hagibis. Together, 2018 and 2019 account for

47% of the nine-year total. Decomposing into fuel and labor clarifies the mechanism. The penalty is borne by labor, which is positive in every year, while fuel registers a net saving over the sample. In the mildest exposure years this fuel saving can exceed the labor penalty, producing a small negative net cost (i.e., savings). In every other year the labor channel dominates. The pooled total is thus a labor-driven figure, partly offset by fuel savings.

Table 2. Annual direct shipping cost penalty, 2013–2021.

Year	Treated voyages	Fuel (\$ M)	Labor (\$ M)	Total excess (\$ M)	Cost premium (%)
2013	5,702	4.6	13.2	17.8	1.4
2014	7,358	-32.1	20.8	-11.3	-0.8
2015	6,160	-8.8	12.0	3.2	0.4
2016	4,434	-3.7	12.8	9.1	1.6
2017	6,059	0.7	18.3	19.0	2.1
2018	9,425	1.4	27.7	29.0	1.8
2019	9,260	-1.2	28.6	27.5	2.0
2020	4,799	-5.4	16.4	11.0	2.0
2021	6,022	-11.5	25.6	14.1	1.4
Total	59,219	-56	175	119	1.2

Notes: Treated voyages are international trips whose least-cost route intersects an active wind swath during the sailing window. Excess cost is realized minus matched-control expected cost from the bunker-normalized voyage panel. Cost penalty is excess over realized.

Why is the fuel-cost effect negative? Exposed voyages slow-steam. Relative to matched controls on the same route, treated voyages travel 5.8% slower and burn 4.2% less fuel per nautical mile, even while covering a slightly longer distance (Table 3). Because fuel use is convex in speed, engine power scales roughly with the cube of speed, so fuel per mile scales with its square. This slowdown more than offsets the detour, lowering fuel cost. The same slowdown keeps vessels at sea 13.6% longer, raising time-related labor cost. The penalty is thus borne by labor, not fuel.

The exposure burden scales with the duration of contact with active wind swaths rather than with their per-voyage intensity. Voyages exposed to the inner 50-knot swath remain in active-swath conditions for 3.9 days on average, roughly twice as long as the 1.9 days experienced by voyages exposed only to the broader 30-knot perimeter. Per-voyage excess cost, however, is similar across the two cohorts (\$2,000 for 50-knot, \$2,100 for 30-knot), which is

Table 3. Slow-steaming mechanism: exposed voyages slow down and burn less fuel per mile, but stay at sea longer.

	Speed (knots) (1)	Duration (hours) (2)	Distance (nm) (3)	Fuel cost per nm (USD) (4)
Typhoon-exposed	-0.7293*** (0.0348)	14.46*** (0.8903)	39.23*** (3.674)	-1.187*** (0.4108)
Observations	441,932	441,932	441,932	441,932
R ²	0.51149	0.91073	0.98848	0.71919
Route fixed effects	✓	✓	✓	✓
Vessel type fixed effects	✓	✓	✓	✓
Tonnage decile fixed effects	✓	✓	✓	✓
Year-month fixed effects	✓	✓	✓	✓

Notes: Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Voyage-level OLS of each outcome on a binary typhoon-exposure indicator, with route, vessel-type, tonnage-decile, and year-month fixed effects. Speed is AIS-track distance over voyage time, and fuel per nm is fuel cost over AIS-track distance. Standard errors clustered at the year-month level.

consistent with the slow-steaming mechanism documented above. That is, at higher intensities vessels throttle back more aggressively, lengthening their voyage duration but reducing fuel consumption per nautical mile. Aggregate cost from 50-knot exposure therefore scales primarily through the longer exposure window, not through a higher per-voyage premium.

These avoidance costs also vary systematically across the fleet. Cargo vessels (bulk carriers and general cargo ships) account for 41% of the total excess cost (\$48 million) and tankers for 69% (\$82 million), with the remainder spread across container ships and specialized vessels. However, the cost penalty as a share of realized voyage cost is higher for tankers (2.1%) than for cargo vessels (0.9%). This difference is consistent with the fleet structure described in Section 2: tankers have higher baseline fuel consumption and, when carrying crude oil or LNG on time-charter contracts, face steeper penalties for delays. Their higher per-voyage cost of avoidance translates into a larger percentage penalty even though their share of total treated voyages is smaller. Appendix A.5 reports the full vessel-type decomposition of treated voyages, excess cost, and percentage penalty.

5.3 The *Ad Valorem Equivalent* of the Typhoon Shock

Before turning to elasticity, we summarize the size of the typhoon shock as an *Ad Valorem Equivalent* (AVE) share of cargo value. Dividing each voyage’s excess cost by the cargo value allocated to that voyage (from monthly port-level customs statistics) yields a weighted-average AVE shock of 0.012% across the \$748 billion of exposed trade over 2013–2021. This is just 1.2 basis points per dollar of trade. The shock is small relative to typical bilateral tariff variation but is concentrated in a handful of port-country-months during the July–October typhoon season, and our IV strategy exploits exactly this within-pair, within-time variation to recover the elasticity. As a long-run cross-sectional benchmark, applying $\varepsilon = -5$ from Head and Mayer (2014) to the bilateral-panel base of \$91 million yields an implied trade-volume loss of \$456 million. Appendix A.3 expands this exercise to alternative elasticities.

We note that this long-run calibration assumes trade adjusts as if the typhoon-induced cost shock were permanent and is therefore an effective upper bound on the welfare loss. In the next section, we will estimate the appropriate short-run elasticity directly.

5.4 Structural Trade Elasticity

Table 4 shows the bilateral trade-cost elasticity to typhoon-induced shocks. Column 1 is the OLS benchmark: a regression of log bilateral imports on $\log(1 + \Delta\tau_{jct})$, where $\Delta\tau_{jct}$ is the realized AVE typhoon shock (excess voyage cost normalized by bilateral trade value). This approach is biased because the regressor is constructed by dividing excess voyage cost by bilateral trade value, and bilateral trade value is itself the dependent variable. A high-trade month thus mechanically yields a small $\Delta\tau_{jct}$ and a low-trade month yields a large one, generating a negative correlation between regressor and outcome by construction.

Columns 2 and 3 address this problem. Column 2 uses a single instrument (the share of voyages on the jc route in month t overlapping any active wind swath). Column 3 uses the intensity decomposition (storm-core 50-knot and 30-knot-only shares jointly). Column 4 strengthens this approach by adding month-of-sample fixed effects to absorb cross-pair

monthly seasonality. Column 5 reports the saturated three-way fixed-effect structure (pair, port-month, country-month). We note that while the latter would be the ideal setup, it pushes identification to its tightest variation and the first-stage F falls just below the conventional weak-instrument threshold.

Table 4. 2SLS estimates of the bilateral trade-cost elasticity.

	OLS (1)	IV: any swath (2)	$\log X_{jct}$ IV: intensity (3)	+ year-month (4)	Saturated (5)
$\log(1 + \Delta\tau_{jct})$	-2.479*** (0.2813)				
$\widehat{\log \bar{c}_{jct}}$ (instrumented)		-1.482** (0.5924)	-1.292** (0.5260)	-2.056* (1.155)	-2.490 (1.815)
Observations	15,448	15,448	15,448	15,448	15,448
R ²	0.84718	0.79332	0.80358	0.76528	0.82621
F-test (1st stage), $\log \bar{c}_{jct}$		42.804	26.386	7.8372	3.8987
Sargan			0.72405	0.00586	0.16127
Port-country pair fixed effects	✓	✓	✓	✓	✓
Year fixed effects	✓	✓	✓		
Year-month fixed effects				✓	
Port $\times$ time fixed effects					✓
Country $\times$ time fixed effects					✓

Notes: Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Dependent variable is $\log X_{jct}$, log bilateral imports from country c through Japanese port j in month t . Column 1 reports OLS on $\log(1 + \Delta\tau_{jct})$, the realized AVE typhoon shock (excess voyage cost divided by bilateral trade value). Columns 2–5 instrument $\log \bar{c}_{jct}$, the log of the mean realized voyage cost on route jc in month t , with typhoon-exposure shares. Column 2 uses the any-swath voyage share as a single instrument. First-stage Kleibergen-Paap F -statistic and Hansen-Sargan over-identification p -value reported below the coefficient (where applicable). Standard errors clustered at the port-country pair $\times$ year level.

Our preferred IV specification (Column 3) yields $\hat{\beta} = -1.292$, implying a trade-cost elasticity of $\widehat{\sigma - 1} \approx 1.29$. The robustness columns line up with the headline: the single-instrument variant (Column 2) yields -1.482 , adding year-month fixed effects (Column 4) leaves the point estimate at -2.056 , and the saturated three-way specification (Column 5) yields -2.490 . Appendix A.4 expands this estimation to a battery of fixed-effect structures and additional instrument decompositions.

We interpret the recovered elasticity as a short-run object, because it is identified from transient weather shocks and captures a margin on which importers delay or re-time ship-

ments rather than permanently abandoning a source country. In other words, this is a partial-adjustment elasticity, the appropriate primitive for evaluating the welfare burden of transient weather shocks.

5.5 Welfare Through Two Channels

We aggregate these results into a two-channel welfare estimate for typhoon exposure on maritime trade in Japanese waters from 2013 to 2021. The first channel is the direct resource cost to the shipping industry: \$119 million in excess fuel and labor across 59,219 treated voyages, identified through the matched-control framework of Section 4.3. Assuming competitive shipping markets, these operational costs are ultimately passed through to consumers via higher freight rates.

The second channel is the structural trade-volume welfare loss. Under the linear iceberg-cost approximation, $\text{welfare}_{\text{struct}} = |\hat{\beta}| \times \text{base}$, where the base represents the aggregate excess shipping costs within the gravity sample. Evaluated at our preferred IV point estimate ($|\hat{\beta}| = 1.29$) on \$91 million of excess shipping cost, this implies an additional \$117.8 million in lost surplus from forgone trade.⁸

The gravity framework naturally magnifies these transport-cost shocks. Across a wide class of CES models, the welfare lost on forgone trade exceeds the extra cost paid to maintain surviving trade, with gains depending strictly on aggregate import shares and the trade elasticity (Arkolakis et al., 2012). Notably, this magnification is moderated by our use of a short-run elasticity: at the long-run Head and Mayer (2014) benchmark ($\varepsilon = -5$), the same base implies a trade-volume loss of \$456 million; this is more than four times our estimate.

The two channels combined imply a total welfare penalty of approximately \$237 million over the nine-year sample. Given the inferential uncertainty inherent in the structural gravity leg, we treat the direct shipping cost as our conservative baseline, as its identification at the

⁸The \$91 million base is smaller than the \$119 million matched-control headline because the bilateral panel restricts to port-country-months with valid customs flows and observed AIS voyages on a matched route; the headline aggregates over the full set of treated voyages.

voyage level is independent of the structural parameters. The two-channel total provides a more complete accounting of the disruption, yielding an estimate that is consistent with the dry-bulk benchmark in Brancaccio et al. (2020). This cumulative burden represents a seasonal economic drain that will likely intensify as climate change alters typhoon climatology in the basin.

6 Discussion

To our knowledge, this is the first voyage-level, fleet-wide estimate of the trade cost of extreme weather, linking a direct shipping-cost channel to a structural welfare loss. Using Japan as a case study, we combined high-resolution wind-swath data with hundreds of thousands of vessel voyages from AIS records. We documented sharp, transient traffic disruptions on exposed routes (up to roughly 40% in open ocean and about 7% at port). We then translated those disruptions into a direct shipping-industry cost of \$119 million through a voyage-level matched-control framework, and used typhoon exposure as a supply-side instrument for bilateral shipping costs. Our estimates suggest an implied \$117.8 million in trade-volume welfare loss. The combined burden, roughly \$237 million over 2013–2021, identifies a cost channel that is invisible to disaster-cost frameworks built around landfall outcomes.

Three caveats temper these estimates. First, the voyage-level cost is estimated on the matched route network, which covers 82% of Japanese-port voyages, 81% of voyage-level operating cost, and 100% of reported customs trade value over 2013–2021 (Appendix A.9). The voyages outside this network are predominantly short-haul, coastal, and ambiguous-origin trips that carry essentially none of Japan’s reported international trade, so the trade-relevant fleet is meaningfully represented. Scaling the headline by the inverse coverage rate, under the assumption that excluded voyages are exposed at the same rate, implies a full-fleet figure of \$147 million. We report this number as a bound and anchor on the matched -sample headline of \$119 million, for which exposure and cost are directly observed.

Second, treatment is defined as any overlap between a voyage’s sailing window and an active wind-swath day, so many voyages coded as treated experienced little actual disruption, attenuating the per-voyage estimate. Third, the matched-control design captures the cost differential relative to a control group that could in principle be exposed to general-equilibrium spillovers (port congestion from delayed treated vessels, bunker-price spikes). To bound this concern, we re-aggregate excess cost using only controls whose entire sailing window is temporally separated by at least 7 days from any typhoon-active day on the same route (Appendix A.8). The spillover-clean excess is \$120 million on 319,000 surviving controls, so we can conclude that within-route spillover contamination is not meaningfully biasing the headline result. Together these mechanisms suggest that our headline result is a conservative one.

The economic burden we identify reveals a quantitatively meaningful, yet historically overlooked, pre-landfall channel that directly informs maritime adaptation policy. Traditional cost-benefit analyses for port infrastructure focus predominantly on physical asset damage and revenue lost during structural repair (Becker et al., 2018; Verschuur et al., 2020). The \$119 million in direct shipping costs we measure accumulated over nearly a decade of ordinary typhoon seasons without a single catastrophic direct hit on a major port complex. Adaptation measures that reduce the necessity of this routine avoidance, such as optimized scheduling systems, have the potential to directly mitigate these operational costs.

The incidence of the costs we measure is itself an open policy question. The shipping industry bears the immediate resource cost; cargo owners and final consumers absorb the pass-through to freight rates; port authorities absorb operating costs from disrupted scheduling and queuing. Each group has distinct adaptation margins and distinct exposure to climate-driven intensification, and the appropriate policy response to a \$237 million nine-year burden differs across them. Furthermore, maritime typhoon risk is currently priced almost exclusively through hull and cargo insurance, paying out only upon physical damage or loss. Voyage operating costs incurred to safely avoid a storm remain uninsurable.

Our disaggregated, voyage-level data provides the actuarial foundation for a novel class of insurance products: seasonal surcharges or voyage-cost-indexed instruments that compensate shipowners for weather-induced operating penalties, functioning similarly to “war risk” premia applied in piracy-affected waters.

The ability to manage these operational costs depends heavily on forecast accuracy. Our event-study results show that vessel traffic declines 24 to 48 hours before a typhoon’s arrival. While not directly observed, this suggests that captains may proactively use forecasts to optimize their exposure; an adaptive margin previously documented in the context of Atlantic hurricanes (Molina and Rudik, 2024). In maritime logistics, the value of a forecast is the cost differential between a planned, early rerouting and a disruptive, last-minute correction.

Looking at this response, we can do a back-of-the-envelope calculation to quantify the value of these forecasts by translating the anticipatory traffic decline into a cost-saving metric. Pre-event adjustments account for 53% of total observed avoidance. If these early diversions had instead been forced into reactive, last-minute exposure, they would have incurred an estimated \$136 million in additional costs over 2013–2021 (roughly \$15 million per year). While this estimate represents an upper bound (i.e., attributing all pre-event movement to forecasts rather than climatological knowledge), it underscores the critical role of information in mitigating climate-driven trade shocks. Given that the underlying data (AIS tracking, wind radii, and fuel prices) are increasingly global, this framework could provide a reproducible template for benchmarking maritime climate adaptation across the world’s most exposed trade corridors.

Finally, while our framework is portable, the historical baseline it captures (2013–2021) is rapidly shifting. Tropical cyclone power dissipation in the Western Pacific has nearly doubled since the 1970s (Emanuel, 2005), driven by a compounding trio of risks: increased storm intensity (Knutson et al., 2020), rising frequencies of rapid intensification (Bhatia et al., 2019), and a slowdown in translation speeds that keeps vessels under extreme conditions for longer periods (Lin and Emanuel, 2016).

Applying these projections to a 2°C warming scenario suggests a meaningful increase in cumulative voyage-day exposure, with the duration component, which is often overlooked relative to intensity, acting as the primary cost driver. Discounted at standard infrastructure rates, this could translate into tens of millions of dollars in present-value costs for the direct channel alone over a 30-year horizon. These figures indicate that the economic penalty we measure is not a static parameter but a climate-sensitive variable. Our findings suggest that long-run maritime adaptation must move beyond landfall-only models to explicitly account for the escalating costs of on-the-water disruption.

7 Conclusion

Extreme weather imposes significant, yet historically unmeasured, costs on global supply chains long before storms make landfall. In this paper, we quantify the economic burden of transient typhoon exposure on maritime trade in Japan. By integrating high-resolution meteorological wind swaths with vessel voyages and bilateral trade flows, we isolate a critical, pre-landfall disruption channel that operates entirely independently of physical infrastructure damage.

Our results reveal that typhoons trigger a significant, yet transient, reorganization of maritime traffic: while vessels at port largely wait out the storm in place, those in the open ocean aggressively reroute or delay, leading to traffic reductions of up to 40%. These disruptions translate into a substantial direct penalty for the shipping industry, driven primarily by increased labor costs and extended voyage durations rather than by additional fuel use, and feed through into welfare losses via reduced trade flows. This underscores that while the opaque cost of storm avoidance is a significant drain on welfare, its magnitude is driven by the immediate operational constraints of the global fleet rather than permanent shifts in trade relationships.

As the climate warms, the frequency of intense tropical cyclones in the Western Pacific

is projected to increase. The economic penalty we document will mechanically scale with this intensification. For economies heavily reliant on maritime logistics, recognizing these routine, on-the-water avoidance costs will be essential. Ultimately, mitigating these growing burdens will require integrating transient weather risks into long-run infrastructure planning, marine insurance pricing, and forecast-optimized maritime routing.

References

- Anderson, James E. and Eric van Wincoop (2003) “Gravity with Gravitas: A Solution to the Border Puzzle,” *American Economic Review*, Vol. 93, No. 1, pp. 170–192.
- Anderson, James and Eric van Wincoop (2004) “Trade Costs,” *Journal of Economic Literature*, Vol. 42, No. 3, pp. 691–751.
- Arkolakis, Costas, Arnaud Costinot, and Andrés Rodríguez-Clare (2012) “New Trade Models, Same Old Gains?” *American Economic Review*, Vol. 102, No. 1, pp. 94–130.
- Asturias, Jose (2020) “Endogenous Transportation Costs,” *European Economic Review*, Vol. 123, p. 103366, Working paper version 2018.
- Barrot, Jean-Noël and Julien Sauvagnat (2016) “Input Specificity and the Propagation of Idiosyncratic Shocks in Production Networks,” *Quarterly Journal of Economics*, Vol. 131, No. 3, pp. 1543–1592.
- Becker, Austin H., Pamela Matson, Michael Fischer, and Michael D. Mastrandrea (2018) “Towards Seaport Resilience for Climate Change Adaptation: Stakeholder Perceptions of Hurricane Impacts in Gulfport (MS) and Providence (RI),” *Progress in Planning*, Vol. 99, pp. 1–49.
- Bensassi, Sami and Inmaculada Martínez-Zarzoso (2013) “The Price of Modern Maritime Piracy,” *Defence and Peace Economics*, Vol. 24, No. 5, pp. 397–418.
- Bernhofen, Daniel M., Zouheir El-Sahli, and Richard Kneller (2016) “Estimating the Effects of the Container Revolution on World Trade,” *Journal of International Economics*, Vol. 98, pp. 36–50.
- Besley, Timothy, Thiemo Fetzer, and Hannes Mueller (2015) “The Welfare Cost of Lawlessness: Evidence from Somali Piracy,” *Journal of the European Economic Association*, Vol. 13, No. 2, pp. 203–239.
- Betz, S. (2011) “Reducing the Risk of Vessel Strikes to Endangered Whales in the Santa Barbara Channel,” Ph.D. dissertation, University of California, Santa Barbara.
- Bhatia, Kieran T., Gabriel A. Vecchi, Thomas R. Knutson, Hiroyuki Murakami, James Kossin, Keith W. Dixon, and Carolyn E. Whitlock (2019) “Recent Increases in Tropical Cyclone Intensification Rates,” *Nature Communications*, Vol. 10, p. 635.
- Board of Governors of the Federal Reserve System (US) (2026) “Japanese Yen to U.S. Dollar Spot Exchange Rate (EXJPUS),” retrieved from FRED, Federal Reserve Bank of St. Louis, Monthly average, not seasonally adjusted.
- Boehm, Christoph E., Aaron Flaaen, and Nitya Pandalai-Nayar (2019) “Input Linkages and the Transmission of Shocks: Firm-Level Evidence from the 2011 Tōhoku Earthquake,” *Review of Economics and Statistics*, Vol. 101, No. 1, pp. 60–75.

- Brancaccio, Giulia, Myrto Kalouptsi, and Theodore Papageorgiou (2020) “Geography, Transportation, and Endogenous Trade Costs,” *Econometrica*, Vol. 88, No. 2, pp. 657–691.
- Bunker Index (2026) “Bunker Index 380 (BIX380): Daily Worldwide Average Price for 380 CST Marine Fuel Oil,” Online data series, <https://www.bunkerindex.com>, Daily price in USD per metric ton.
- Carvalho, Vasco M., Makoto Nirei, Yukiko U. Saito, and Alireza Tahbaz-Salehi (2021) “Supply Chain Disruptions: Evidence from the Great East Japan Earthquake,” *Quarterly Journal of Economics*, Vol. 136, No. 2, pp. 1255–1321.
- Conley, T. G. (1999) “GMM Estimation with Cross Sectional Dependence,” *Journal of Econometrics*, Vol. 92, No. 1, pp. 1–45.
- Corbett, J. J., H. Wang, and J. J. Winebrake (2009) “The effectiveness and costs of speed reductions on emissions from international shipping,” *Transp. Res. D Transp. Environ.*, Vol. 14, pp. 593–598.
- Dell, Melissa, Benjamin F. Jones, and Benjamin A. Olken (2014) “What Do We Learn from the Weather? The New Climate-Economy Literature,” *Journal of Economic Literature*, Vol. 52, No. 3, pp. 740–798.
- Deryugina, Tatyana (2017) “The Fiscal Cost of Hurricanes: Disaster Aid Versus Social Insurance,” *American Economic Journal: Economic Policy*, Vol. 9, No. 3, pp. 168–198.
- Emanuel, Kerry (2005) “Increasing Destructiveness of Tropical Cyclones over the Past 30 Years,” *Nature*, Vol. 436, No. 7051, pp. 686–688.
- Englander, Gabriel (2019) “Property Rights and the Protection of Global Marine Resources,” *Nature Sustainability*, Vol. 2, No. 10, pp. 981–987.
- Felbermayr, Gabriel and Jasmin Gröschl (2014) “Naturally Negative: The Growth Effects of Natural Disasters,” *Journal of Development Economics*, Vol. 111, pp. 92–106.
- Feyrer, James (2021) “Distance, Trade, and Income—The 1967 to 1975 Closing of the Suez Canal as a Natural Experiment,” *Journal of Development Economics*, Vol. 153, p. 102708.
- Global Fishing Watch (2021) “Anchorages, ports and voyages data,” <https://globalfishingwatch.org/datasets-and-code-anchorages/>.
- (2026) “Anchorages Version 2,” <https://globalfishingwatch.org/data-download/datasets/public-anchorages%3Av20200316>, Data set accessed 2026-04-06.
- Head, Keith and Thierry Mayer (2014) “Gravity Equations: Workhorse, Toolkit, and Cookbook,” in Gita Gopinath, Elhanan Helpman, and Kenneth Rogoff eds. *Handbook of International Economics*, Vol. 4: Elsevier, pp. 131–195.

- Hsiang, Solomon M. and Amir S. Jina (2014) “The Causal Effect of Environmental Catastrophe on Long-Run Economic Growth: Evidence from 6,700 Cyclones,” NBER Working Paper 20352, National Bureau of Economic Research.
- Huang, Zhucheng, Zhangyuan He, Pengjun Zhao, Caixia Zhang, Yingnan Niu, Wuxin Guo, Yanzhe Cui, and Weizeng Shao (2024) “The effects of tropical cyclone on the container shipping network: A case study of typhoon Ma-on (2022),” *Transportation Research Part D: Transport and Environment*, Vol. 136, p. 104449.
- Hummels, David (2007) “Transportation Costs and International Trade in the Second Era of Globalization,” *Journal of Economic Perspectives*, Vol. 21, No. 3, pp. 131–154.
- Hummels, David L and Georg Schaur (2013) “Time as a trade barrier,” *Am. Econ. Rev.*, Vol. 103, No. 7, pp. 2935–2959.
- IEA (2021) “Japan 2021: Energy Policy Review,” Technical report, International Energy Agency, Paris.
- International Maritime Organization (2000) “SOLAS Regulation V/19: Carriage Requirements for Shipborne Navigational Systems and Equipment,” International Convention for the Safety of Life at Sea (SOLAS), 1974, as amended; Chapter V, AIS carriage requirements adopted 2000, entered into force 2002.
- International Transport Workers’ Federation (2025) “ITF Seafarers – wages and compensation,” <https://www.itfseafarers.org/en/resources/wages>. Accessed: 2025-11-06.
- Kitamoto, Asanobu, Jared Hwang, Bastien Vuillod, Lucas Gautier, Yingtao Tian, and Tarin Clanuwat (2023) “Digital Typhoon: Long-term Satellite Image Dataset for the Spatio-Temporal Modeling of Tropical Cyclones,” in *Advances in Neural Information Processing Systems 36 (NeurIPS Datasets and Benchmarks Track)*.
- Knutson, Thomas, Suzana J. Camargo, Johnny C. L. Chan, Kerry Emanuel, Chang-Hoi Ho, James Kossin, Mrutyunjay Mohapatra, Masaki Satoh, Masato Sugi, Kevin Walsh, and Liguang Wu (2020) “Tropical Cyclones and Climate Change Assessment: Part II: Projected Response to Anthropogenic Warming,” *Bulletin of the American Meteorological Society*, Vol. 101, No. 3, pp. E303–E322.
- Krichene, Hazem, Thomas Vogt, Franziska Piontek, Tobias Geiger, Christof Schötz, and Christian Otto (2023) “The social costs of tropical cyclones,” *Nature Communications*, Vol. 14, No. 1, p. 7294.
- Kroodsma, D. A., J. Mayorga, T. Hochberg, N. A. Miller, K. Boerder, F. Ferretti, A. Wilson, B. Bergman, T. D. White, B. A. Block et al. (2018) “Tracking the global footprint of fisheries,” *Science*, Vol. 359, pp. 904–908.
- Limão, Nuno and Anthony J. Venables (2001) “Infrastructure, Geographical Disadvantage, Transport Costs, and Trade,” *World Bank Economic Review*, Vol. 15, No. 3, pp. 451–479.

- Lin, Ning and Kerry Emanuel (2016) “Grey Swan Tropical Cyclones,” *Nature Climate Change*, Vol. 6, pp. 106–111.
- MAFF (2022) “Japan’s Food Domestic Production Ratio and Feed Self-Sufficiency Ratio, FY2021,” Technical report, Ministry of Agriculture, Forestry and Fisheries of Japan, Tokyo.
- Ministry of Land, Infrastructure, Transport and Tourism (2014) “National Land Numerical Information: Port and Harbour Data (C02), 2014,” Government of Japan, National Land Numerical Information download service, Shapefile C02-14-g_PortAndHarbor.
- MLIT (2022) “White Paper on Land, Infrastructure, Transport and Tourism in Japan, 2022,” Technical report, Ministry of Land, Infrastructure, Transport and Tourism of Japan, Tokyo.
- Molina, Renato and Ivan Rudik (2024) “The Social Value of Hurricane Forecasts,” NBER Working Paper 32548, National Bureau of Economic Research.
- Molina, Renato, Juan Carlos Villaseñor-Derbez, Gavin McDonald, and Grant R. McDermott (2024) “Dangerous Waters: The Economic Toll of Piracy on Maritime Shipping,” Working Paper SSRN 4811789, University of Miami.
- Mori, Nobuhito, Tomohiro Yasuda, Taro Arikawa, Tomoya Kataoka, Sota Nakajo, Kojiro Suzuki, Yusuke Yamanaka, and Adrean Webb (2019) “2018 Typhoon Jebi Post-Event Survey of Coastal Damage in the Kansai Region, Japan,” *Coastal Engineering Journal*, Vol. 61, No. 3, pp. 278–294.
- Pelli, Martino and Jérôme Tschopp (2017) “Comparative Advantage, Capital Destruction, and Hurricanes,” *Journal of International Economics*, Vol. 108, pp. 315–337.
- Sampson, Charles R., Edward M. Fukada, John A. Knaff, Brian R. Strahl, Michael J. Brennan, and Timothy Marchok (2017) “Tropical Cyclone Gale Wind Radii Estimates for the Western North Pacific,” *Weather and Forecasting*, Vol. 32, No. 3, pp. 1029–1040.
- Schötz, Christof (2025) “Spatial Correlation in Economic Analysis of Climate Change,” *Nature*, Vol. 644, No. 8076, pp. E27–E30.
- Schreck III, Carl J., Kenneth R. Knapp, and James P. Kossin (2014) “The Impact of Best Track Discrepancies on Global Tropical Cyclone Climatologies using IBTrACS,” *Monthly Weather Review*, Vol. 142, No. 10, pp. 3881–3899.
- Stopford, Martin (2009) *Maritime Economics*, London: Routledge, 3rd edition.
- Strobl, Eric (2011) “The Economic Growth Impact of Hurricanes: Evidence from US Coastal Counties,” *Review of Economics and Statistics*, Vol. 93, No. 2, pp. 575–589.
- Sun, Liyang and Sarah Abraham (2021) “Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects,” *Journal of Econometrics*, Vol. 225, No. 2, pp. 175–199.

- UNCTAD (2021) “Review of Maritime Transport 2021,” Technical report, United Nations Conference on Trade and Development, Geneva.
- Verschuur, Jasper, Elco E. Koks, and Jim W. Hall (2020) “Port disruptions due to natural disasters: Insights into port and logistics resilience,” *Transportation Research Part D: Transport and Environment*, Vol. 85, p. 102393.
- Vespe, Michele, Harm Greidanus, and Marlene Alvarez Alvarez (2015) “The declining impact of piracy on maritime transport in the Indian Ocean: Statistical analysis of 5-year vessel tracking data,” *Marine Policy*, Vol. 59, pp. 9–15.
- Wang, C., J. J. Corbett, and J. Firestone (2007) “Modeling energy use and emissions from North American shipping: application of the ship traffic, energy, and environment model,” *Environ. Sci. Technol.*, Vol. 41, pp. 3226–3232.
- Wong, Woan Foong (2022) “The Round Trip Effect: Endogenous Transport Costs and International Trade,” *American Economic Journal: Applied Economics*, Vol. 14, No. 4, pp. 127–166, Working paper version 2019.
- Zhang, Caixia, Pengjun Zhao, Zhucheng Huang, Zhangyuan He, Yingnan Niu, Guangyu Huang, and Yuting Chen (2024) “Assessing vessel transportation delays affected by tropical cyclones using AIS data and a Bayesian network: A case study of Veronica in north-western Australia,” *Ocean Engineering*, Vol. 308, p. 118385.

Appendix

A.1 Unit cost estimation

Pooling across the 536,380 valid voyages and 19,598 vessels in the 2013–2021 panel, fuel cost averages \$38.71 per nautical mile and labor cost averages \$3,354 per day. Both figures are ratios of sums. That is, total cost across all voyage-miles or voyage-days in the panel divided by the corresponding total exposure instead of per-voyage means or medians. In other words, estimates are not influenced by a small set of voyages with extreme cost-per-mile or cost-per-day ratios.

Two qualitative patterns are worth noting. First, fuel cost per nautical mile rises sharply with vessel tonnage, from under \$10 per nautical mile in the smallest deciles to roughly \$50 per nautical mile in the top decile, consistent with marine-engineering predictions on hull-resistance scaling. Second, year-to-year variation in fuel cost per nautical mile tracks the global bunker-price cycle closely, while labor cost per day is essentially flat across the sample period once vessel-mix composition is held constant. Table A1 reports the tonnage-decile breakdown.

Table A1. Unit costs by vessel tonnage decile, 2013–2021.

Tonnage decile	Voyages	Fuel (\$/nm)	Labor (\$/day)
1	1,409	10.97	2,005
2	12,026	9.51	2,875
3	34,382	9.41	3,296
4	49,435	9.82	3,115
5	50,625	14.06	3,198
6	75,245	20.40	3,184
7	97,017	30.75	3,192
8	52,195	28.39	3,260
9	48,222	30.55	3,194
10	115,824	53.10	3,541

Notes: Voyage panel as in Section 3. Tonnage deciles are constructed on gross tonnage across the full voyage sample, with decile 1 the lightest and decile 10 the heaviest. Fuel and labor unit costs are ratios of sums within each decile: total fuel (labor) cost divided by total voyage-nautical-miles (voyage-days). Voyage counts are the underlying sample sizes used to compute each ratio.

A.2 Matching quality

Of the 59,219 treated voyages in the estimation sample, 76% are matched at the tightest stratum (route $\times$ vessel type $\times$ year), 20% at the intermediate stratum (route $\times$ vessel type), 2% at the coarsest stratum (route only), and 1% sit in cells with fewer than five controls at every level and are assigned a route-mean fallback. Fewer than 0.2% of treated voyages have no usable control at any level and are dropped before the final aggregation.

Table A2 reports covariate balance between treated and untreated voyages using the standardized mean difference (SMD). Unconditional balance is poor by design ($\max |\text{SMD}| = 0.77$): treated voyages are mechanically longer and on heavier vessels because longer, heavier voyages have more chances to overlap with an active wind swath. The relevant diagnostic is balance *within* the matching stratum. Residualizing each covariate on stratum fixed effects (route $\times$ vessel type $\times$ year, restricted to strata with both treated and untreated voyages) drops the maximum SMD to 0.24 and leaves only 3 of six covariates with $|\text{SMD}| > 0.1$. Of those, the largest residual is on voyage time itself, where treated voyages are slightly longer than within-stratum controls. This is the expected slow-steaming/avoidance signature and an outcome margin rather than a confounder. Optimal-route distance and tonnage decile are essentially zero by construction (route and vessel type enter the stratum). Climatological wind and wave height residuals are small in absolute terms and reverse sign once stratum is partialled out, indicating that treatment is driven by typhoon *events* rather than by systematically rougher baseline conditions.

Table A2. Covariate balance between treated and untreated voyages.

Covariate	Mean		Std. mean diff.	
	Treated ($N = 59,326$)	Untreated ($N = 382,606$)	Pooled	Within-stratum
Realized distance (nm)	2844	1328	+0.76	+0.09
Voyage time (hours)	233	107	+0.77	+0.17
Optimal-route distance (nm)	2696	1239	+0.74	+0.00
Tonnage decile (1–10)	8.10	6.73	+0.63	+0.01
Climatological wind speed (m/s)	2.93	2.76	+0.15	-0.24
Climatological wave height (m)	1.53	1.42	+0.27	-0.12

Notes: Treated voyages are those whose sailing window intersects an active 30/50-knot wind swath polygon. “Pooled” standardized mean differences (SMDs) use unconditional treated and untreated means. “Within-stratum” SMDs are computed on residuals from a regression of each covariate on route $\times$ vessel type $\times$ year fixed effects, restricted to strata containing both treated and untreated voyages. Climatological wind speed and wave height are voyage-route averages from reanalysis baselines, not realized in-voyage conditions.

A.3 Sensitivity to trade elasticity

Our preferred trade-volume welfare estimate uses the IV-2SLS elasticity $\widehat{\sigma - 1} \approx 1.29$ from Section 4.4. Evaluated on the \$91 million of matched-control excess shipping cost in the bilateral panel, this implies a trade-volume welfare loss of \$117.8 million. Because the elasticity is identified with appreciable noise, Table A3 reports welfare under a range of alternatives: (i) our point estimate with ± 1 Wald SE, (ii) the saturated-FE robustness estimate from Section 4.4, (iii) the Brancaccio et al. (2020) and Wong (2022) IV benchmarks, and (iv) the long-run Head and Mayer (2014) cross-sectional range.

Across the IV scenarios that bound the data, structural welfare spans \$70–227 million and total nine-year welfare spans \$189–346 million. The Head–Mayer long-run benchmarks lie above this range; we report them for completeness but treat them as upper bounds that would apply only if typhoon-induced cost shocks were permanent. Our preferred IV estimate sits within one standard error of the Brancaccio et al. (2020) dry-bulk benchmark, supporting the reading of $\widehat{\sigma - 1} \approx 1.29$ as a credible short-run partial-adjustment elasticity.

Table A3. Welfare sensitivity to the trade-cost elasticity.

Scenario	$\sigma - 1$	Structural welfare (\$M)	Total welfare (\$M)	Note
Preferred IV -1 SE	0.77	70	189	Lower one-SE band.
Preferred IV (pair + year)	1.29	118	237	Headline.
Preferred IV $+1$ SE	1.82	165	284	Upper one-SE band.
Saturated IV (pair, port-month, country-month)	2.49	227	346	Saturated multilateral resistance.
BKP (2020), dry bulk	1.03	94	213	Tariff IV, LIML.
Wong (2019), container	3.00	273	392	Round-trip Bartik IV.
HM (2014), $\varepsilon = -3$	3.00	273	392	Cross-sectional, lower.
HM (2014), $\varepsilon = -5$	5.00	455	574	Cross-sectional, mid.
HM (2014), $\varepsilon = -7$	7.00	637	756	Cross-sectional, upper.

Notes: Structural welfare equals $|\sigma - 1| \times \$91$ million, the linear iceberg-cost approximation evaluated on the bilateral-panel cost base. Total welfare adds the matched-control direct shipping cost of \$119 million. “Preferred IV” is the pair-plus-year FE intensity 2-IV estimate; “ ± 1 SE” uses the cluster-robust Wald SE. “Saturated IV” is the structural-gravity FE specification with pair, port-month, and country-month FE. “BKP (2020), dry bulk” is from Brancaccio et al. (2020). “Wong (2019), container” is from Wong (2022). “HM (2014)” calibrations are the long-run cross-sectional range from Head and Mayer (2014).

A.4 IV robustness: fixed effects, sample, and instrument decomposition

Table A4 evaluates our trade elasticity under eight different fixed-effect structures to ensure our results are not driven by specific modeling choices. The most aggressive saturated specification (Column 5) controls for nearly all simultaneous shifts in global supply and demand, though this approach leaves very little residual variation for the instruments to exploit, causing the first-stage F -statistic to fall slightly. To remain transparent in this high-control environment, we report Anderson–Rubin confidence sets, which provide reliable inference even when the statistical link between typhoon exposure and shipping costs is relatively thin.

Our preferred specification (Pair + Year) strikes a balance, providing a strong statistical foundation (first-stage $F \approx 29$, well above the conventional weak-instrument threshold) while yielding stable point estimates. The results remain consistent across various intermediate structures, and over-identification tests confirm that both our high-intensity and low-intensity storm measures identify a common parameter.

The final rows of Table A4 address a primary identification concern: that typhoon timing (July–October) might simply coincide with the natural seasonal cycle of Asian trade. To test this, we include monthly controls to absorb recurring seasonal demand shocks. While this naturally attenuates our estimate, as it removes the very seasonal variation upon which typhoon identification relies, the effect remains meaningful and the sign remains correct. Crucially, even under these strict seasonal controls, our results stay consistent with the dry-bulk benchmark of -1.03 documented in Brancaccio et al. (2020).

Ultimately, our headline specification represents a deliberate trade-off that prioritizes a strong statistical signal. We therefore treat the seasonality-adjusted figures as a robust lower bound for our findings.

Table A5 examines whether our results are sensitive to the volume of trade on a given route. While our main identification relies on the 15,448 bilateral cells with matched AIS

voyages, we re-estimate the elasticity after restricting the sample to “thick” pairs with at least 50, 100, or 200 voyages over the sample period. The point estimate remains negative under the ≥ 50 and ≥ 100 voyage filters (roughly -0.85 to -0.95), modestly attenuated relative to the -1.17 baseline but of the same sign and order of magnitude. The strictest ≥ 200 filter renders the estimate uninformative (a small, wrong-signed $+0.13$ with a standard error several times its magnitude): even though the first stage remains adequate ($F \approx 14$), restricting to the most heavily traded pairs throws away the within-pair variation in typhoon exposure on which our identification depends.

Table A6 explores the sensitivity of our results to how typhoon exposure is defined. We consider four instrument constructions: a simple aggregate (any-swath share), an intensity decomposition (50-knot vs. 30-knot), a geometry decomposition (core vs. flank), and all four jointly. Across all specifications, the point estimates remain stable within the range of $[-1.201, -1.482]$, and Sargan p -values confirm the mutual validity of these different measures. The point estimates are statistically indistinguishable across the aggregate, intensity, and geometry instrument constructions, and the over-identification tests are never rejected. This stability indicates that our results are driven by the fundamental transport disruption of the typhoon itself, rather than the specific way the wind swath is categorized.

Finally, we address the possibility that typhoon exposure shifts the composition of vessels rather than the cost of the voyages themselves. Our headline regressor is a simple average that could, in principle, reflect changes in the vessel-type mix from month to month. To isolate the pure cost channel, we construct an index that holds each pair’s vessel-type mix constant at its 2013–2014 base-period shares. Re-running our preferred specification on this fixed-basket index yields an estimate of $\hat{\beta} = -1.342$, which is statistically indistinguishable from our headline result (Table A7). This consistency confirms that the observed trade response is driven by the heightened costs of maritime transport rather than a shift in the underlying fleet composition.

Table A4. IV robustness: 2SLS elasticity across fixed-effect structures.

Fixed-effect structure	$\hat{\beta}$	SE	First-stage F	Sargan p	AR 95% CI	n
Saturated (pair + port-month + country-month)	-2.490	(1.815)	3.90	0.69	[−10.00, 0.50]	15,448
pair + port-month	-2.436*	(1.301)	7.05	0.85	[−10.00, −0.10]	15,448
pair + country-month	-1.536	(1.203)	6.45	0.70	[−10.00, 0.90]	15,448
pair + year-month	-2.056*	(1.155)	7.84	0.94	[−10.00, 0.10]	15,448
pair + port-quarter + country-quarter	-0.884	(0.671)	13.46	0.52	[−3.40, 0.50]	15,448
Preferred (pair + year)	-1.292**	(0.526)	26.39	0.39	[−2.90, −0.30]	15,448
pair + year + calendar-month	-0.981	(0.679)	14.59	0.89	[−3.40, 0.50]	15,448
pair $\times$ calendar-month	-0.454	(0.626)	15.21	0.29	[−2.30, 0.90]	15,448

Notes: Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Each row is a 2SLS estimate of equation (4) with $\log \bar{\tau}_{jct}$ instrumented by the 50-knot and 30-knot-only exposure shares, using the fixed-effect structure indicated. Standard errors clustered at the port-country pair $\times$ year level. “First-stage F ” is the Kleibergen-Paap rank statistic. “Sargan p ” is the Hansen-Sargan over-identification p -value (NA when the specification is just-identified).

Table A5. IV robustness: sample sensitivity (preferred FE, intensity 2-IV).

Sample	$\hat{\beta}$	SE	First-stage F	Sargan p	n
Baseline (any voyages)	-1.292**	(0.526)	26.39	0.39	15,448
Pair total voyages ≥ 50	-1.080*	(0.565)	22.11	0.27	14,122
Pair total voyages ≥ 100	-0.818	(0.598)	17.90	0.03	11,808
Pair total voyages ≥ 200	+0.092	(0.600)	13.44	0.04	9,258

Notes: Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Each row re-estimates the preferred (pair plus year FE, intensity 2-IV) specification on the indicated subsample. “Pair total voyages” is the cumulative voyage count on a port-country pair over 2013–2021. Standard errors clustered at the pair $\times$ year level.

Table A6. IV robustness: alternative instrument decompositions (preferred FE).

Instrument set	$\hat{\beta}$	SE	First-stage F	Sargan p	n
Single (any-swath share)	-1.482**	(0.592)	42.80	—	15,448
Intensity (50-knot, 30-only)	-1.292**	(0.526)	26.39	0.39	15,448
Geometry (storm core, flank-only)	-1.468**	(0.590)	21.43	0.45	15,448
All four IVs	-1.201**	(0.509)	18.21	0.59	15,448

Notes: Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Each row uses a different set of typhoon-exposure instruments under the preferred pair plus year FE. “Single” uses the any-swath share. “Intensity” decomposes by storm intensity (50-knot vs 30-knot-only). “Geometry” decomposes by storm geometry (storm core vs flank-only). “All four IVs” uses both decompositions jointly. In the “All four IVs” specification, the flank-only instrument is dropped for collinearity with the intensity and core instruments; that row is identified by the remaining three. Standard errors clustered at the pair $\times$ year level.

Table A7. IV robustness: fixed-basket cost index (vessel-type mix held constant).

	$\log X_{jct}$		
	Headline IV	Headline IV (basket sample)	Fixed-basket IV
	(1)	(2)	(3)
$\widehat{\log \bar{c}_{jct}}$	-1.292** (0.5260)	-1.377** (0.5486)	
$\widehat{\log \bar{c}_{jct}^{\text{basket}}}$			-1.342** (0.5241)
Observations	15,448	15,151	15,151
R ²	0.80358	0.79889	0.80399
F-test (1st stage), $\log \bar{c}_{jct}$	26.386	25.354	
F-test (1st stage), $\log \bar{c}_{jct}^{\text{basket}}$			28.488
Sargan	0.72405	1.3139	1.1491
Port-country pair fixed effects	✓	✓	✓
Year fixed effects	✓	✓	✓

Notes: Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Column 1 reproduces the preferred IV on the full IV sample. Column 2 restricts to the basket-eligible sample (cells with at least one vessel-type stratum whose pair has base-period weight from 2013–2014) and re-estimates the headline. Column 3 replaces the within-cell mean cost with the Laspeyres fixed-basket index: for each pair-month cell, the cost is the base-period vessel-type-weighted average of stratum-mean costs, with weights renormalized over strata realized in the cell. Standard errors clustered at the pair $\times$ year level.

A.5 Vessel heterogeneity

In this section, we explore how the burden of typhoon exposure varies across different segments of the maritime fleet. Table A9 reports the matched-control excess costs by vessel type, revealing that while tankers and bulk carriers absorb the highest costs in absolute terms, the relative cost premium is highest for container and reefer vessels (roughly 3%) and lowest for bulk cargo (roughly 1%). This divergence aligns with the operational rigidity of different shipping markets: container liners operating on fixed schedules face steep penalties for delays, whereas bulk carriers on flexible “tramp” schedules can absorb weather-driven disruptions at a lower marginal cost. We treat negative cost estimates in long-tail categories with limited samples as statistical noise rather than realized savings.

We corroborate this schedule-rigidity hypothesis parametrically in Table A8. The results show that bulk cargo carries a small, statistically insignificant penalty (\$600 per voyage). In contrast, tankers and container/reefer vessels incur penalties that are an order of magnitude larger and significant at the 1% level (\$8,800 and \$3,200, respectively). This ordering confirms that the economic cost of storm avoidance is fundamentally determined by a vessel’s operational flexibility, with schedule-rigid segments bearing the brunt of the disruption.⁹

⁹The matched-control premiums and the regression coefficients answer different counterfactual questions and use different denominators; however, both yield a consistent qualitative ordering that identifies bulk cargo as the most flexible and schedule-rigid vessels as the least.

Table A8. Voyage-level treatment effects on total cost, by vessel-type group.

	Total cost (USD)		
	Cargo (1)	Tanker (2)	Container/Reefer (3)
Typhoon-exposed	563.8 (379.1)	8,775.9*** (1,176.5)	3,204.4*** (606.3)
Observations	323,245	94,522	14,274
R ²	0.86902	0.85608	0.92663
Route fixed effects	✓	✓	✓
Tonnage decile fixed effects	✓	✓	✓
Year-month fixed effects	✓	✓	✓

Notes: Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Each column estimates $\text{Cost}_{ijt} = \beta \text{Treated}_{ijt} + \alpha_j + \alpha_d + \alpha_{ym} + \varepsilon_{ijt}$ on voyages of the corresponding vessel-type group, where α_j is a route fixed effect, α_d a tonnage-decile fixed effect, and α_{ym} a year-month fixed effect. Container/Reefer pools the four reefer-family classes from Table A9 (container reefer, reefer, specialized reefer, and cargo or reefer). Standard errors clustered at the route level.

Table A9. Direct shipping cost of typhoon exposure, by vessel type.

Vessel type	Treated voyages	Excess cost (\$ M)	Cost premium (%)
Tanker	14,524	82.2	2.1
Cargo	41,639	48.3	0.9
Container Reefer	1,030	3.2	3.5
Cargo or Reefer	359	0.5	1.3
Reefer	205	0.0	0.1
Bunker	6	0.0	1.8
Bunker or Tanker	44	-0.2	-9.7
Specialized Reefer	522	-0.3	-0.8
Cargo or Tanker	890	-14.5	-14.6
Total	59,219	119	1.2

Notes: Excess cost is computed using the matched-control framework described in Section 5. Vessel-type labels follow the AIS-derived best-vessel-type classification reported by Global Fishing Watch; combined labels (e.g., “Cargo or Tanker”) are vessels whose registry information is ambiguous between two parent classes. Cost premium is excess cost divided by realized total cost on treated voyages. Vessel-types with fewer than 100 treated voyages should be interpreted as descriptive only.

A.6 Spatial patterns of typhoon exposure

In this section, we present the spatial substrate that the cell-level event study from Section 5 averages over. Figure A1 reports the count of cell-days under any active wind swath (30- or 50-knot, short or long side) between 2013 and 2021 on the 0.5° EEZ grid. From this figure, note that exposure concentrates along the southern and central Pacific approaches to Japan, with a clear secondary band over the Philippine Sea where storms recurve before transiting toward the archipelago. We interpret this as descriptive evidence consistent with the event-study results in the main text: the busiest shipping corridors out of Japan are also the corridors most heavily traversed by recurring typhoons.

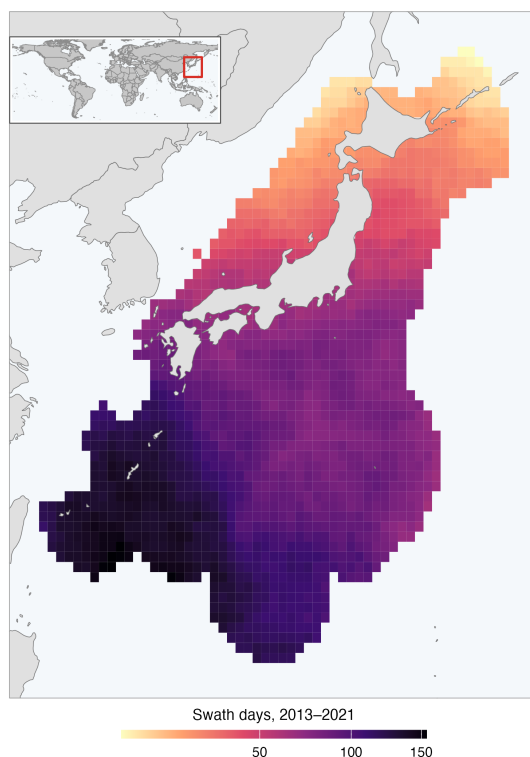


Figure A1. Geography of typhoon wind-swath exposure on the 0.5° EEZ grid. The unit of observation is a 0.5° cell-day. Each tile shows the count of cell-days with any active wind swath (30-knot short side, 30-knot long side, 50-knot short side, or 50-knot long side) over 2013–2021, with a square-root color scale.

A.7 Replication with the 50-knot wind core

Here we rerun our entire pipeline with treatment restricted to the 50-knot wind core. This is the inner region of the typhoon where mariners universally prioritize avoidance. This restriction serves as a conservative robustness check to ensure our headline results are not driven by the broader, less intense 30-knot perimeter.

The qualitative patterns remain unchanged under this stricter threshold. Grid-cell and port-level event studies (Figures A2 and A3) show traffic reductions that are larger in absolute value than the headline, consistent with more aggressive avoidance under intense winds. Crucially, the cost composition reveals the “slow-steaming” mechanism in its most extreme form: at the 50-knot threshold, vessels throttle back so aggressively that they actually realize fuel savings (\$44 million) while incurring substantial labor penalties (\$113 million) from extended voyage durations. This fuel-versus-time tradeoff is confirmed by the voyage-level regressions in Table A11.

The structural trade response also remains remarkably stable. Instrumenting bilateral costs with the 50-knot exposure share yields an elasticity of $\widehat{\sigma - 1} \approx 1.15$, which is statistically indistinguishable from our headline estimate (Table A12). While certainly noisy, this consistency confirms that the short-run trade-cost elasticity is a fundamental response to maritime transport disruption, regardless of the specific wind intensity used to define the shock.

Combining these channels yields a total welfare penalty of roughly \$122 million over the sample period. While this is approximately half of our central \$237 million estimate, the reduction is a mechanical consequence of the 50-knot core covering a smaller spatial and temporal footprint than the full 30-knot swath. We read this exercise as evidence that our headline mechanism is robust to how extreme weather exposure is operationalized.

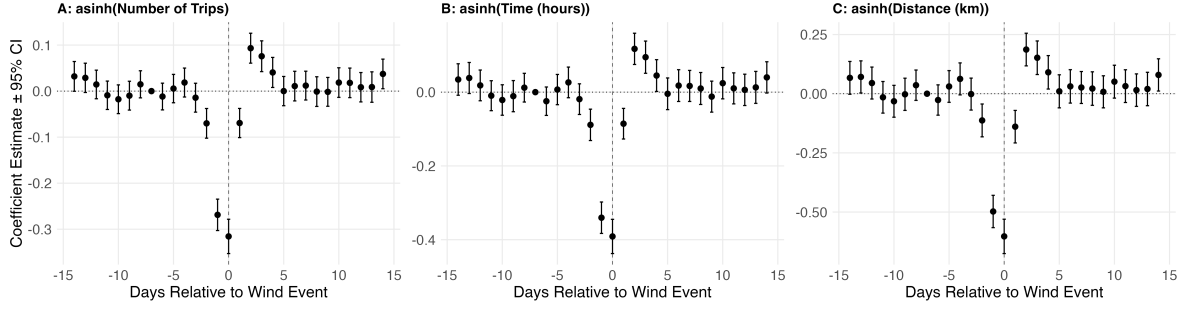


Figure A2. Grid-cell event study, 50-knot threshold. Construction, treatment definition, and standard errors as in Figure 3, except the treatment indicator is $1[\text{wind_knot_mean}_{it} \geq 50]$.

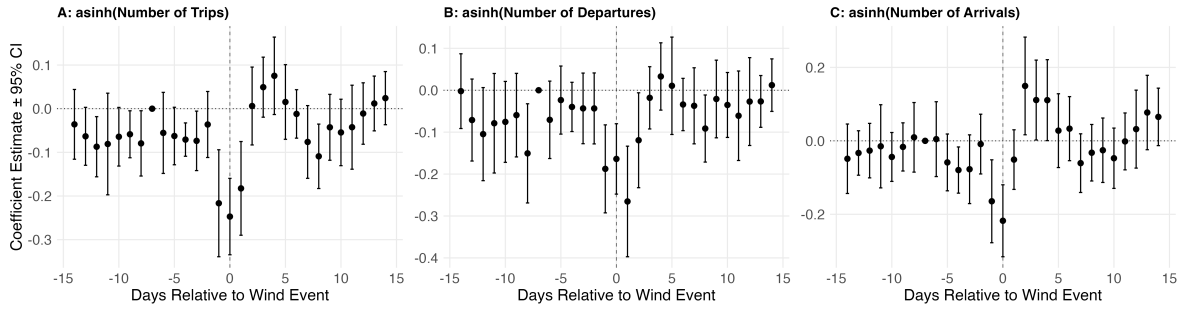


Figure A3. Port-level event study, 50-knot threshold. Sample and specification as in Figure 4, except the treatment indicator activates only when an active 50-knot swath polygon covers the port.

Table A10. Annual direct shipping cost penalty, 50-knot threshold, 2013–2021.

Year	Treated voyages	Excess cost (\$ M)	Cost premium (%)
2013	2,622	2.9	0.4
2014	3,056	7.2	1.0
2015	3,463	3.7	0.7
2016	2,695	7.7	1.9
2017	2,929	8.5	1.8
2018	4,873	16.3	1.8
2019	3,742	13.3	2.1
2020	2,101	8.3	3.2
2021	1,814	1.2	0.3
Total	27,295	69	1.4

Notes: As in Table 2, except a voyage is treated only when its sailing window intersects an active 50-knot swath (vs the 30/50 union in the headline).

Table A11. Voyage-level treatment effects on operating cost, 50-knot threshold.

	Fuel cost (USD)	Labor cost (USD)	Total cost (USD)
	Fuel cost (1)	Labor cost (2)	Total cost (3)
Typhoon-exposed (50-knot core)	-157.4 (650.1)	2,914.5*** (112.2)	2,757.1*** (650.9)
Observations	441,932	441,932	441,932
R ²	0.81603	0.88887	0.85510
Route fixed effects	✓	✓	✓
Vessel type fixed effects	✓	✓	✓
Tonnage decile fixed effects	✓	✓	✓
Year-month fixed effects	✓	✓	✓

Notes: Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. As in Table 1, except the binary treatment indicator activates only when a voyage's sailing window intersects an active 50-knot swath polygon.

Table A12. Structural gravity estimates, 50-knot threshold.

	log X_{jct}	
	Preferred FE (1)	Saturated FE (2)
$\widehat{\log \bar{c}_{jct}}$	-1.150** (0.5451)	-2.321 (1.774)
Observations	15,448	15,448
R ²	0.81046	0.83556
F-test (1st stage), log $\bar{c}_{jct}$	46.670	6.9314
Port-country pair fixed effects	✓	✓
Year fixed effects	✓	
Port $\times$ time fixed effects		✓
Country $\times$ time fixed effects		✓

Notes: Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. As in Table 4, except the exposure share used as the instrument is the share of voyages overlapping a 50-knot wind swath only. The first-stage Kleibergen-Paap F -statistic appears below the coefficient.

A.8 Bound on matched-control direct cost

The matched-control design assumes that untreated voyages provide a valid counterfactual for those exposed to typhoons. A potential concern is that these controls may be subject to general-equilibrium spillovers, such as port congestion or scheduling cascades, stemming from concurrent disruptions on the same route. If such spillovers increase the costs for untreated vessels, our measured excess cost would be attenuated.

We bound this potential bias by identifying a subset of “spillover-clean” controls. We flag any untreated voyage as being at risk if its sailing window, expanded by a ± 7 -day buffer, overlaps with any typhoon activity on its route. Removing these at-risk observations leaves 319,000 clean controls (roughly 83% of the original pool). Re-estimating the direct costs using only this clean sample yields an excess cost of \$120 million, or 101% of our headline figure. Because these estimates are statistically indistinguishable, we conclude that within-route spillovers do not meaningfully bias our results. Our primary matched-control estimate thus remains a conservative and reliable anchor for the total burden.

Table A13 confirms the clean estimate is not driven by a particular buffer choice. We sweep the buffer width from ± 3 to ± 14 days; the surviving control pool falls monotonically with the buffer, but the recovered clean excess and its share of the headline are essentially flat across the range, leaving the spillover-bound conclusion unchanged.

Table A13. Spillover bound on direct cost: sensitivity to the buffer window.

Buffer	Clean controls	Clean excess (\$ M)	% of headline
± 3 days	349,609	113	95
± 5 days	332,514	120	101
± 7 days (headline)	318,866	120	101
± 10 days	302,046	111	93
± 14 days	284,935	114	96

Notes: Each row re-runs the matched-control aggregation using only untreated voyages whose sailing window, expanded by the indicated $\pm$ buffer in days, does not overlap any typhoon-active day on the same route. “Clean controls” is the surviving untreated pool. “Clean excess” is the total excess shipping cost recomputed on the clean controls, holding the L1/L2/L3 fallback stratification of Section 4.3 fixed. “% of headline” is the clean excess divided by the matched-control headline of \$119 million.

A.9 Coverage of the matched-route IV sample

Our matched-control and structural estimates are identified using a network of primary Japanese port units. Here we quantify the representativeness of this network relative to the broader Japanese maritime aggregate from 2013 to 2021.

As shown in Table A14, our matched sample captures the vast majority of Japan’s maritime activity, accounting for 82% of total voyages, 81% of aggregate vessel operating costs, and 100% of total customs trade value. Under the assumption that excluded voyages face similar exposure patterns (because they traverse the same Western Pacific corridors) scaling our headline direct cost by the inverse coverage rate would imply a full-fleet burden of \$147 million. While we provide this figure as a transparency bound on the total disruption, we anchor our main analysis to the matched-sample headline (\$119 million) where exposure and costs can be directly verified.

Table A14. Coverage audit: matched-route IV sample vs Japanese trade aggregate, 2013–2021.

Dimension	In matched network	Outside matched network
Voyages (2013–2021)	441,932 (82%)	94,448 (18%)
Operating cost (\$M)	35,013 (81%)	8,194 (19%)
Customs trade (¥ trillion)	836 (100%)	0 (0%)

Notes: Voyage panel restricted to international trips departing from or arriving at Japanese ports between 2013 and 2021 with finite fuel and labor cost. Operating cost is the sum of voyage-level fuel and labor cost. Customs trade is the sum of import and export values across all reporting Japanese ports over the same window. “In matched network” rows reflect voyages, cost, and customs flow associated with the international ports underlying our IV identification; “Outside” rows are the residual.